



Biologists create mutant mice

Biologists have created mice that live longer and appear to age slower than ordinary mice. <https://digit.in/jun18-B31-1a>



Double Trojan Horse Drug

The drug fools bacteria into thinking its nutritious and releases the second drug inside the bacteria to kill it. <https://digit.in/jun18-B31-2>

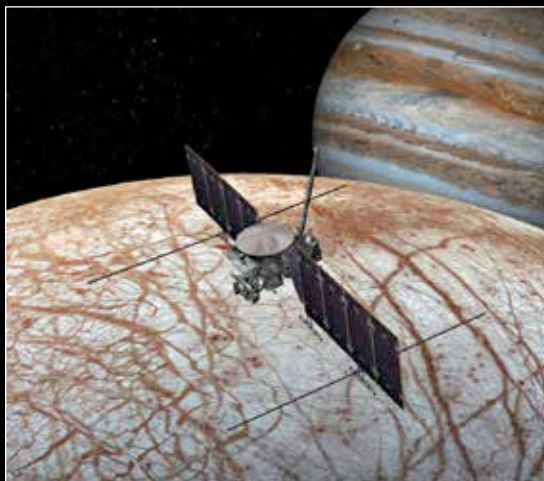
MARS

The first pathfinding mission to Mars, to identify potential landing spots for the first colonial ship for the SpaceX base was scheduled to launch in 2018 itself. This was to be accomplished by a Dragon spacecraft, which was going to the Red Planet, which was why the mission was called the Red Dragon mission. Looks like

the mission has been rescheduled. The InSight mission to Mars which launched this month is expected to reach its destination in November. Then, Mars will receive a health checkup. A seismograph will act like a stethoscope, and measure the pulse of the planet, or vibrations in the ground. The instrument is sensitive enough to pick up vibrations due to storms. A self digging nail will carry a ribbon

loaded with temperature sensors, to measure the temperature of the interior of Mars. We know a lot about the surface and the atmosphere, but very little about what lies beneath the surface of the Red Planet. The year 2020 is really the breakout year for Mars explorations. There are just too many missions scheduled for that year. An as yet unnamed Mars rover by NASA, and the Exomars rover by ESA will both be looking for signs of past and present life. The United Arab Emirates will send out its Hope orbiter. China aims for its maiden mission to Mars, and will send both a rover and an orbiter. India plans its second mission to Mars, Mangalyaan-2 in either 2020 or 2021. After that, there are only proposals for missions, no concrete plans. ESA and JAXA both have incipient plans for sample return missions to Phobos in the mid 2020s. NASA's PADME orbiter is a mission to dedicatedly study the moons of Mars. The JAXA Melos mission to Mars has a rover loaded with an aircraft. The BOLD mission by NASA plans

to use six impact landers to study the surface for signs of ancient or current life. The next decade is going to be very exciting indeed, when it comes to the exploration of Mars. SpaceX is the only entity ambitious enough to have plans for a manned mission to Mars within the next decade. There are some other private



A concept rendering of the Europa Clipper mission

spaceflight companies trying to get there first, Inspiration Mars and Mars One, but both of these projects are highly unlikely.

THE SOLAR SYSTEM

There are two high profile scheduled missions to the inner Solar System in the next ten years. The BepiColombo mission to Mercury and the Parker Solar Probe to the Sun. Only two probes have visited the innermost planet so far, Mariner 10 and Messenger. The BepiColombo mission is made up of two probes, that will both be inserted in orbit around Mercury. The mission is expected to take off later this year from Earth, will arrive at Mercury in 2021, and will continue to study the planet till 2027 or 2028. The probe is equipped to study high energy particles around Mercury.

The Parker Solar Probe will fly closer to the Sun than any other spacecraft before it. In fact, it will fly right into the corona of the Sun. The delicate scientific instruments on board are protected by a 4.5 inch

thick reinforced carbon-carbon composite shield. The spacecraft is designed to withstand temperatures of 1,377 degrees Celsius. The Parker Solar Probe will face more heat and radiation than any other spacecraft so far. The Parker Solar Probe is also scheduled for launch later this year, and will arrive at the Sun in 2024.

There are planned missions to the outer Solar System as well. The Europa Clipper mission by NASA is expected to take off between 2022 and 2025. Its mission is to study Europa, map the subsurface ocean, and if possible, find signs of life. The Europa Clipper is considered a follow up mission to the Galileo probe. Data gathered from the Galileo probe, and the Hubble Space Telescope have indicated that there are jets of water erupting from Europa. These jets give direct access to the subsurface ocean. Scientists previously believed that it would be required to drill through the ice cap and dispatch submarine drones to study the subsurface ocean on Europa. The jets change everything, because a probe with the right sensors on it can directly sample the material in the ocean, by flying through the jets. Galileo is believed to have flown through one of these geysers in a flyby in 1997, but unfortunately did not have the sensors on board to analyse the particles. The Europa Clipper mission is equipped with the requisite sensors. The JUPITER ICy moons Explorer (JUICE) mission by ESA will study Ganymede, Callisto and Europa. All three moons have global subsurface oceans. The probe will map the surfaces of all the three moons, and peek into the water layers beneath. It will also study the atmosphere on Ganymede, as well as the interaction of its magnetic field with the Jovian magnetosphere. JUICE is scheduled for launch in 2022. The observation campaigns of Europa Clipper and JUICE are expected to overlap. If the subsurface oceans of these icy moons harbour life, we may know of it within the next ten years. ■